Advanced Classical Mechanics

Lecture 7: Conservation of Linear Momentum

Readings: Morin - Chapter 5, Marion Chapter 2

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The Law of Conservation of Linear Momentum

The **law of conservation of linear momentum** states that the sum of the momenta of the components of an isolated system (on which no force acts) remains constant in time.

The **center-of-momentum** frame (**COM** frame) or **zero-momentum** frame is the inertial frame within which the total linear momentum of the system under study is zero:

$$\sum_{i=1}^{N} \boldsymbol{P}_i = \sum_{i=1}^{N} m\boldsymbol{v}_i = 0$$

- In a center-of-momentum frame, the system's center of mass is at rest but not necessarily at the origin of the coordinate system.
- The energy of the system is minimal in a center-of-momentum frame.
- The **center-of-mass frame (CM)** is a special case of the center-of-momentum frame, defined as an inertial frame whose **origin** always corresponds to **the center of mass of the system** under study:

$$\sum_{i=1}^{N} \boldsymbol{P}_i = \sum_{i=1}^{N} m\boldsymbol{v}_i = 0$$

$$\sum_{i=1}^{N} m\boldsymbol{r}_i = 0$$

- Solving Classical Mechanics problems in CM frames is often desired as it significantly simplifies the solutions. The default frame within which problems are frequently defined is frequently called the **lab frame**, which is not necessarily the same as the CM frame. Solving problems often involves switching back and forth between the lab and CM frames.

The Law of Conservation of Linear Momentum

Problem 1. (Linear momentum conservation from Newton's laws) While the law of conservation of linear momentum sounds like an axiom, it can also be derived from Newton's laws. Consider an isolated system comprised of two interacting bodies that obey the Newtonian dynamics. Show that the total momentum of this system must be constant over time.

Problem 2. (Snow on a sled): You are riding on a sled that is given an initial push and slides across frictionless ice. Snow is falling vertically (in the frame of the ice) on the sled. Assume that the sled travels in tracks that constrain it to move in a straight line. Which of the following three strategies causes the sled to move the fastest? The slowest?

A: You sweep the snow off the sled so that it leaves the sled in the direction perpendicular to the sled's tracks, as seen by you in the frame of the sled.

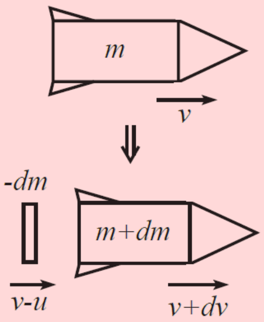
B: You sweep the snow off the sled so that it leaves the sled in the direction perpendicular to the sled's tracks, as seen by someone in the frame of the ice.

C: You do nothing.

Problem 3. (Rocket Motion): Most of a rocket's mass consists of fuel, which is eventually ejected by the rocket, pushing the rocket forward based on the law of the conservation of momentum.

Consider a simple rocket that ejects a mass dm backward with a (constant) speed u relative to the rocket. Since u is speed, it is positive by definition.

Let the rocket have initial mass M and let m be the changing mass at a later time.



a) Show that the change in velocity between the first moment when the rocket has mass m_1 and speed v_1 and a later moment when the rocket has a mass m_2 and velocity v_2 is,

$$\Delta v = v_2 - v_1 = u \log\left(\frac{m_1}{m_2}\right)$$

b) The above speedup is dismal for a rocket. How can you build a rocket that goes faster than this?

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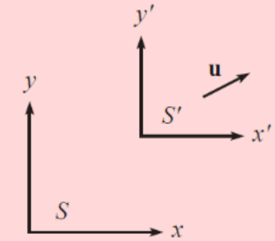
Problem 4. (Center of Mass (CM) Frame)
Consider a system of N particles each with mass m_i and location $\boldsymbol{r}_i$ moving with speed $\boldsymbol{v}_i$ with the index i running from 1 to N in the coordinate system S.
Derive the corresponding equations for the velocity $\boldsymbol{u}$ and origin $\boldsymbol{r}_{com}$ of the center-of-mass inertial frame S' for this system in terms of the m_i, $\boldsymbol{r}_i$, and $\boldsymbol{v}_i$.



Problem 5. (Two masses in 1-D) A mass m with speed v approaches a stationary mass M. The masses bounce off each other without any loss in total energy. What are the final velocities of the particles? Assume that the motion takes place in one dimension.

Problem 6. (Minimum kinetic energy) The total kinetic energy of the system of particles in Problem 4 in reference frame S is defined as,

$$T = \frac{1}{2}\sum_{i=1}^{N} m_i v_i^2$$

Show that the kinetic energy of the system of particles defined in Problem 4 is minimum only in a COM frame.

Problem 7. (Conservation of relative velocity) An **elastic collision** (in an arbitrary number of dimensions) is a collision that **conserves the total kinetic energy** of the system before and after the collision.
Consider a mass m with speed v approaching a stationary mass M.
The masses bounce off each other elastically.
Using the notion of elastic collisions,
what are the final velocities of the particles?
Assume that the motion takes place in one dimension.

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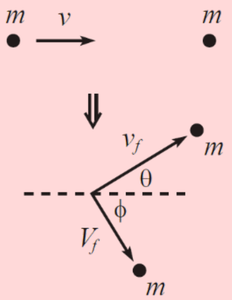
Problem 8. (Conservation of relative velocity)
Show that in a 1-D elastic collision, the relative velocity of the two particles after the collision is the negative of the relative velocity before the collision.

Problem 9. (Billiards)
A billiard ball with speed v approaches an identical stationary one.
The balls bounce off each other elastically, deflecting the incoming ball by an angle θ.



a) What are the final speeds of the balls?
b) What is the angle φ at which the stationary ball is deflected?

Problem 10. (Inherently inelastic processes: Sand on conveyor belt)
Sand drops vertically (from a negligible height) at a rate σ kg/s onto a moving conveyor belt.
a) What force must you apply to the belt to keep it moving at a constant speed v?
b) How much kinetic energy does the sand gain per unit time?
c) How much work do you do per unit time?
d) How much energy is lost to heat per unit time?

Problem 11. (Chain on a scale) A completely flexible, infinitesimally thin, and non-stretchable chain with length L and mass density σkg/m is held such that it hangs vertically just above a scale. It is then released. What is the reading on the scale as a function of the height of the top of the chain?

Problem 12. (Drag force on a sphere) A sphere of mass M and radius R moves with speed V through a region of space that contains particles of mass m that are at rest with a volumetric density n per unit volume. Assume $m \ll M$, and assume that the particles do not interact with each other. What is the effective drag force experienced by the sphere?